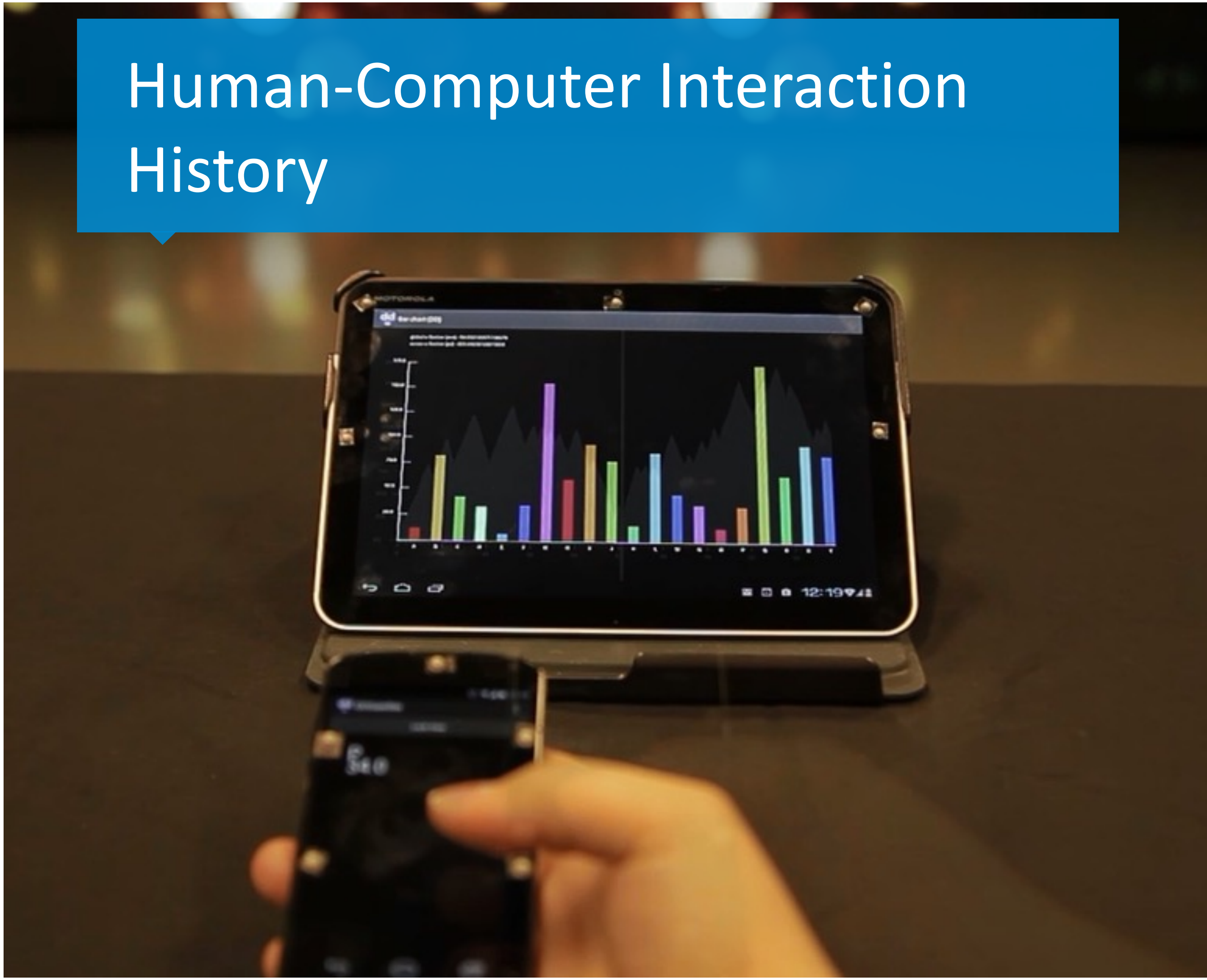
Human-Computer Interaction History



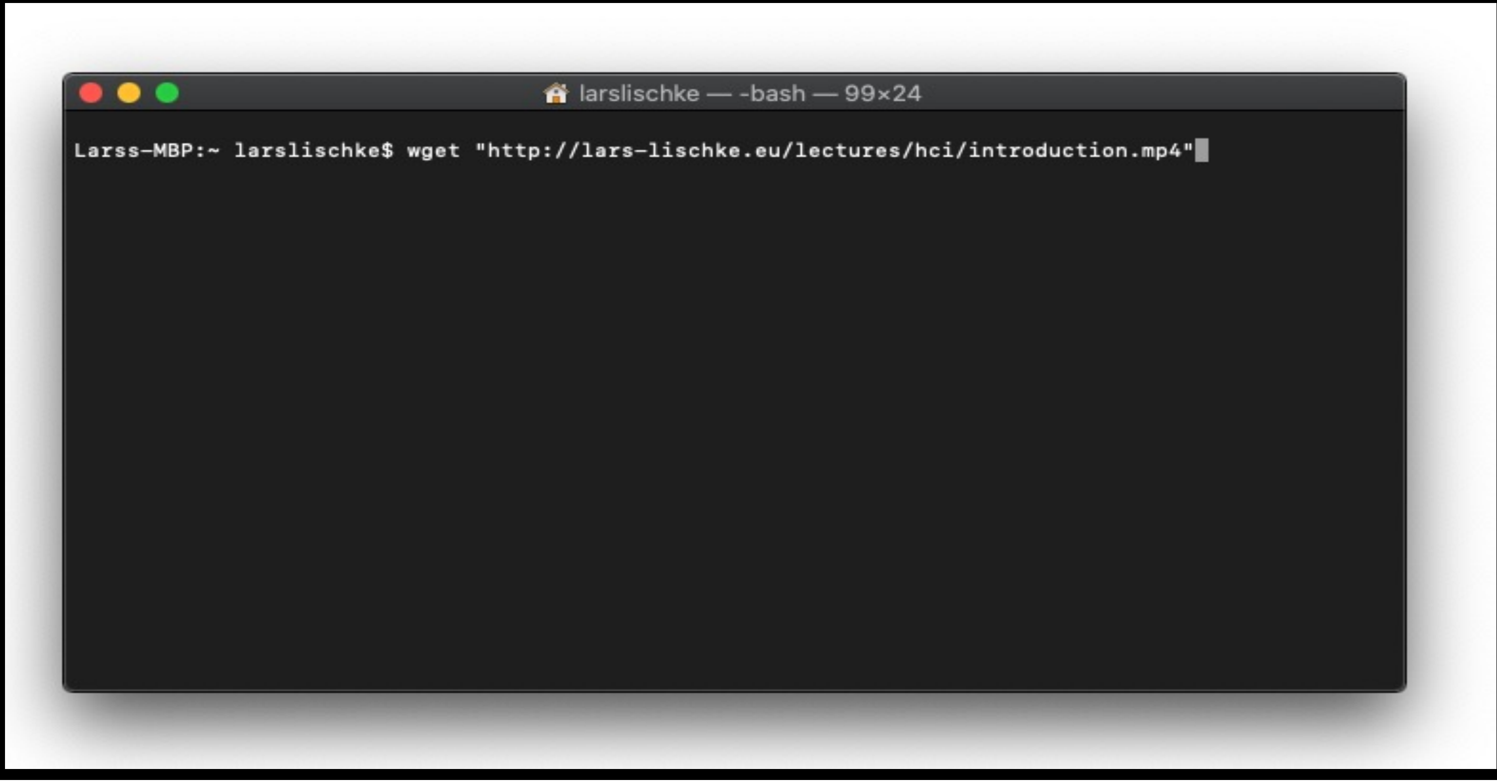
Human-Computer Interaction

“Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them”
(working definition in the ACM SIGCHI Curricula for HCI [1])

[1] Thomas T. Hewett, Ronald Baecker, Stuart Card, Tom Carey, Jean Gasen, Marilyn Mantei, Gary Perlman, Gary Strong, and William Verplank. 1992. *ACM SIGCHI Curricula for Human-Computer Interaction*. Technical Report. ACM, New York, NY, USA.

When did Human-Computer Interaction become important?

- We have computers since the 1940s…
- Interaction studies since the 1980s…



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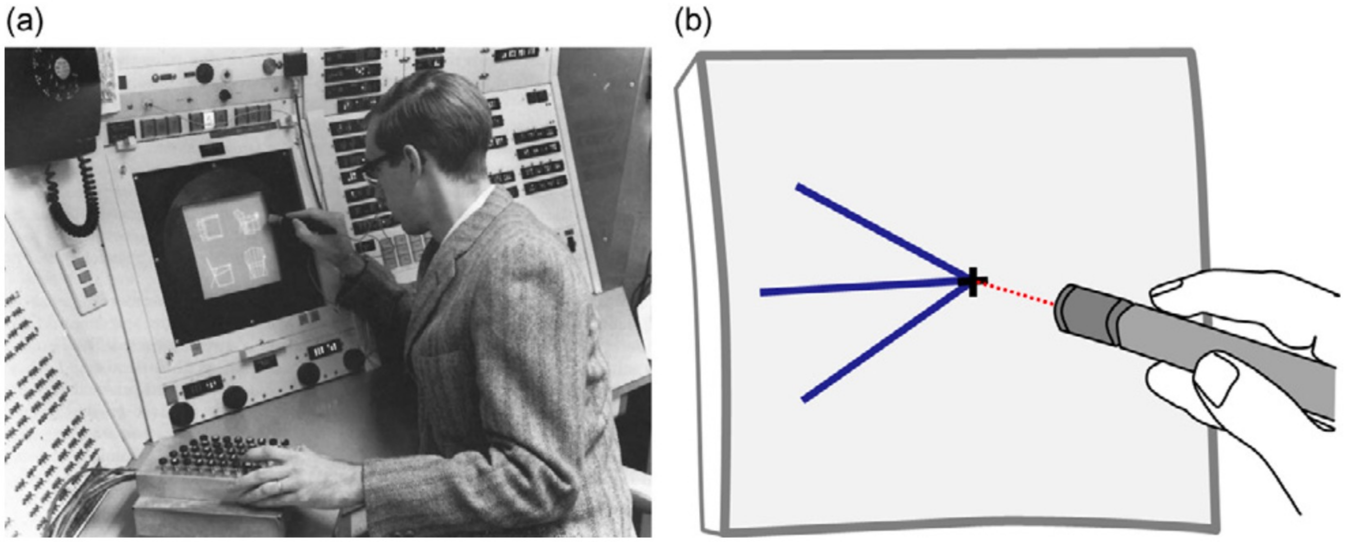

FIGURE 1.3

(a) Demo of Ivan Sutherland's Sketchpad. (b) A light pen dragging ("rubber banding") lines, subject to constraints.

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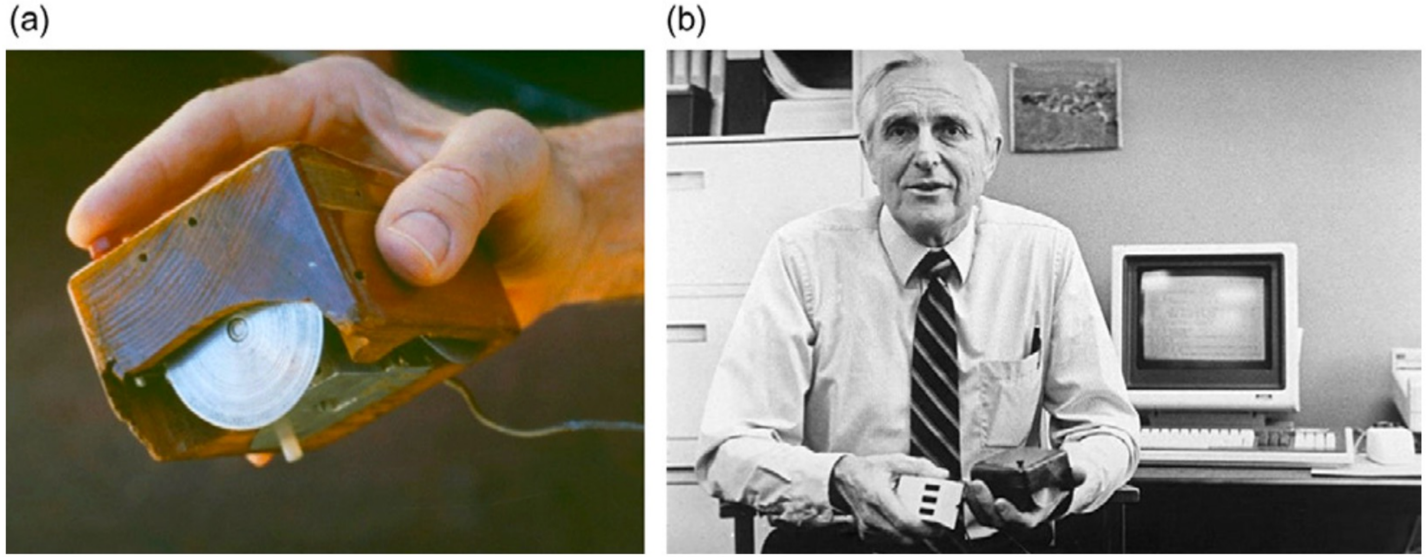


FIGURE 1.4

(a) The first mouse. (b) Inventor Douglas Engelbart holding his invention in his left hand and an early three-button variation in his right hand.

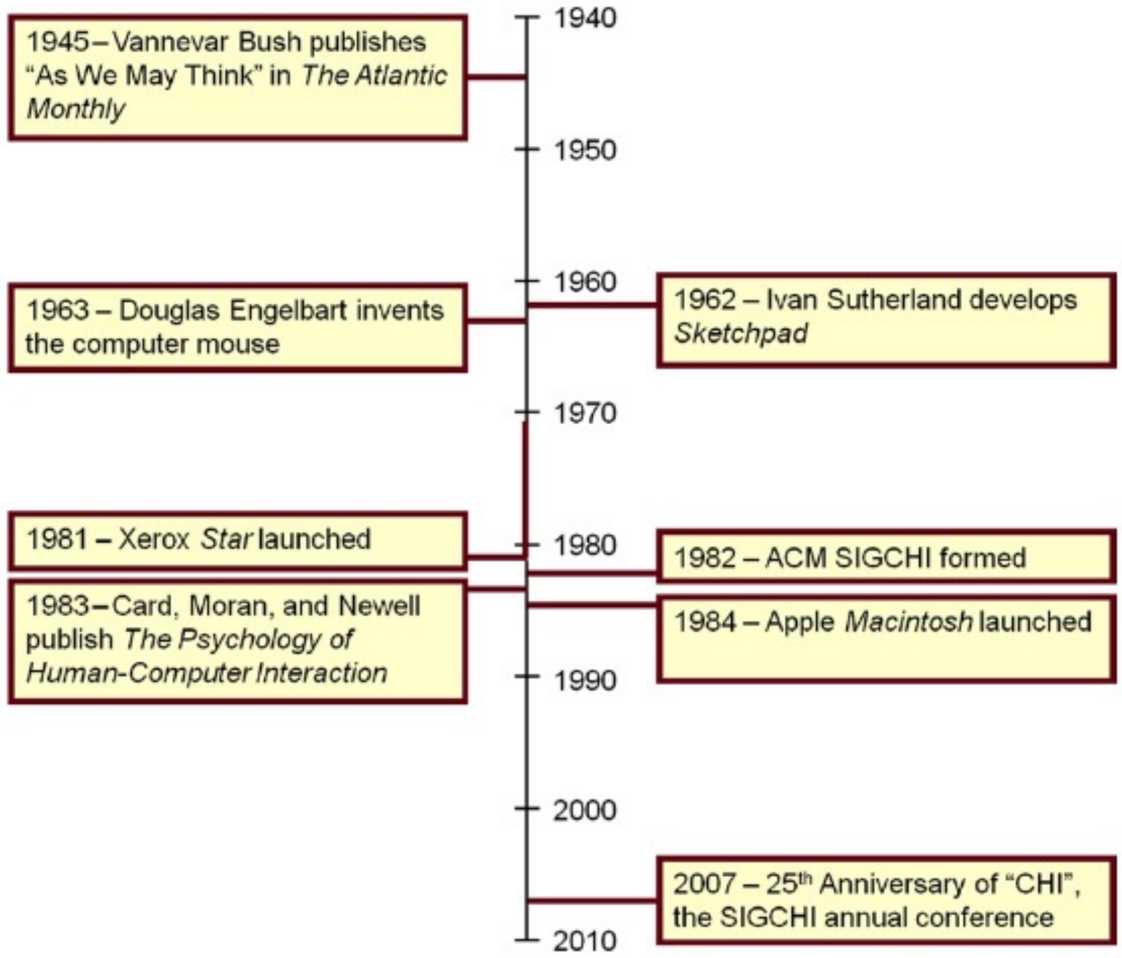


FIGURE 1.1

Timeline of notable events in the history of human–computer interaction HCI.

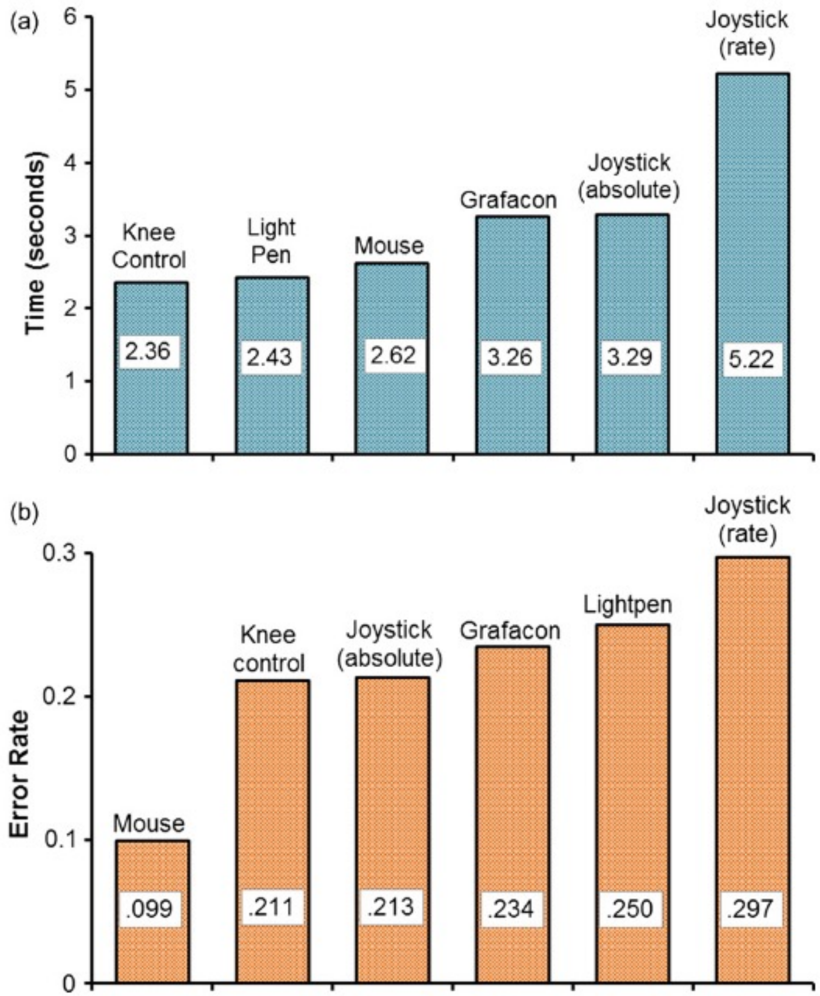


FIGURE 1.6

Results of first comparative evaluation of a computer mouse: (a) Task completion time in seconds. (b) Error rate as the ratio of missed selections to all selections.

(Adapted from English et al., 1967)

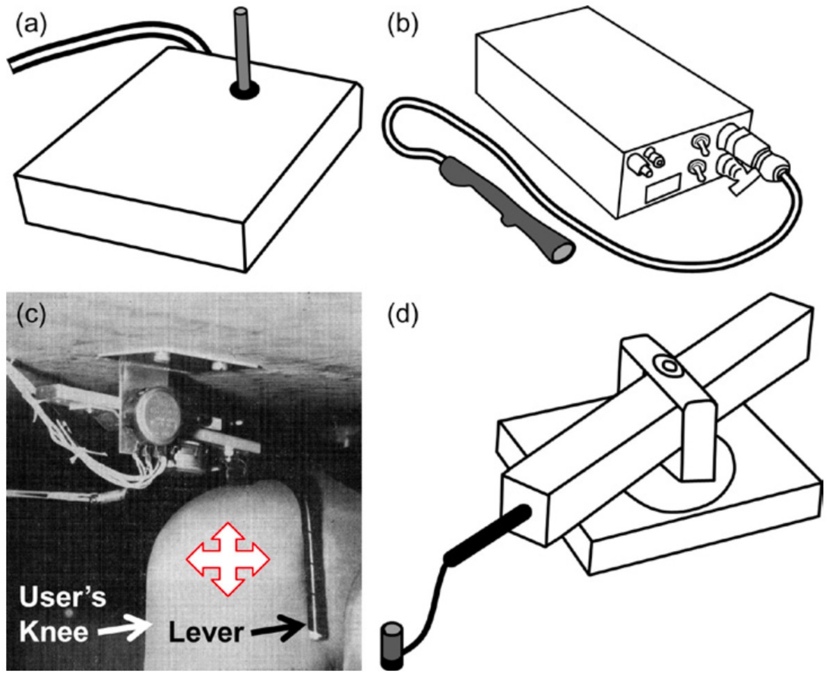


FIGURE 1.5

Additional devices used in the first comparative evaluation of a mouse: (a) Joystick. (b) Lightpen. (c) Knee-controlled lever. (d) *Grafacon*.

(1967) Mouse evaluation experiment by English

First commercial computer with a GUI (1981)

FIGURE 1.7

Xerox Star workstation.

Also the first to use the “*desktop*” metaphor

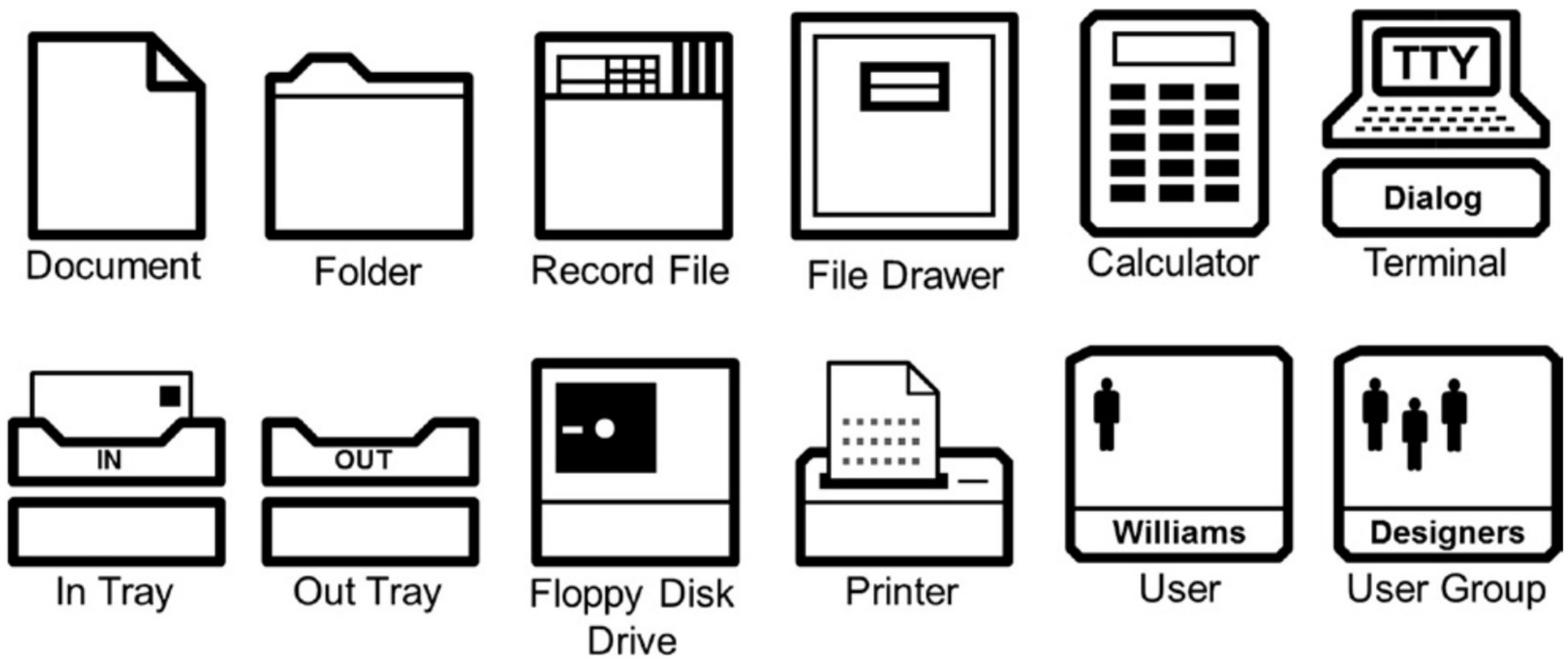


FIGURE 1.8

Examples of icons appearing on the Xerox Star desktop.

Birth of HCI discipline
(first ACM CHI conference in 1983)

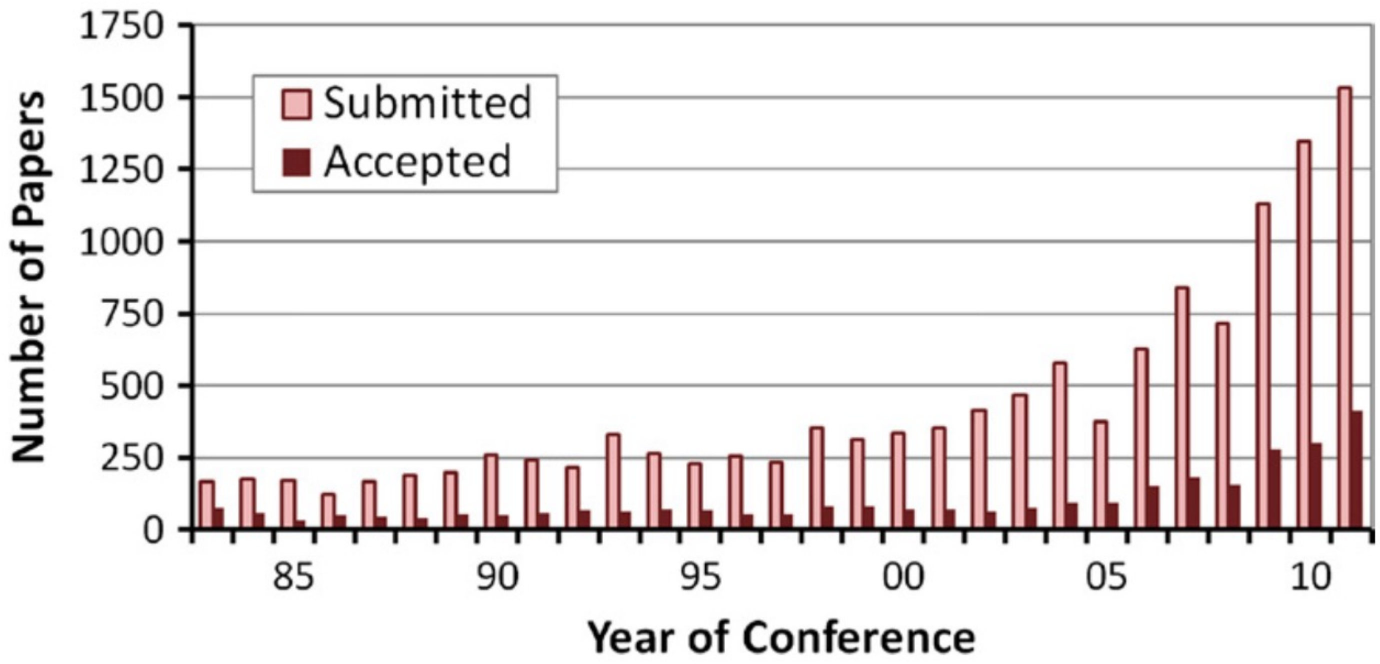


FIGURE 1.9

Number of papers submitted and accepted by year for the ACM SIGCHI Conference on Human Factors in Computing Systems ("CHI"). Statistics from the ACM Digital Library.

The *model human processor* (MHP)

A cognitive modeling method developed by Card, Moran & Newell (1983) used to calculate how long it takes to perform a certain task.

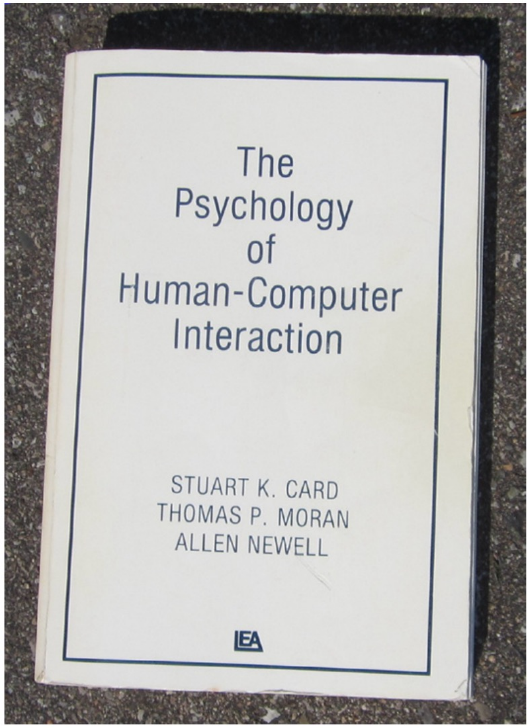


FIGURE 1.10

Card, Moran, and Newell's *The Psychology of Human-Computer Interaction.*

(Published by Erlbaum in 1983)

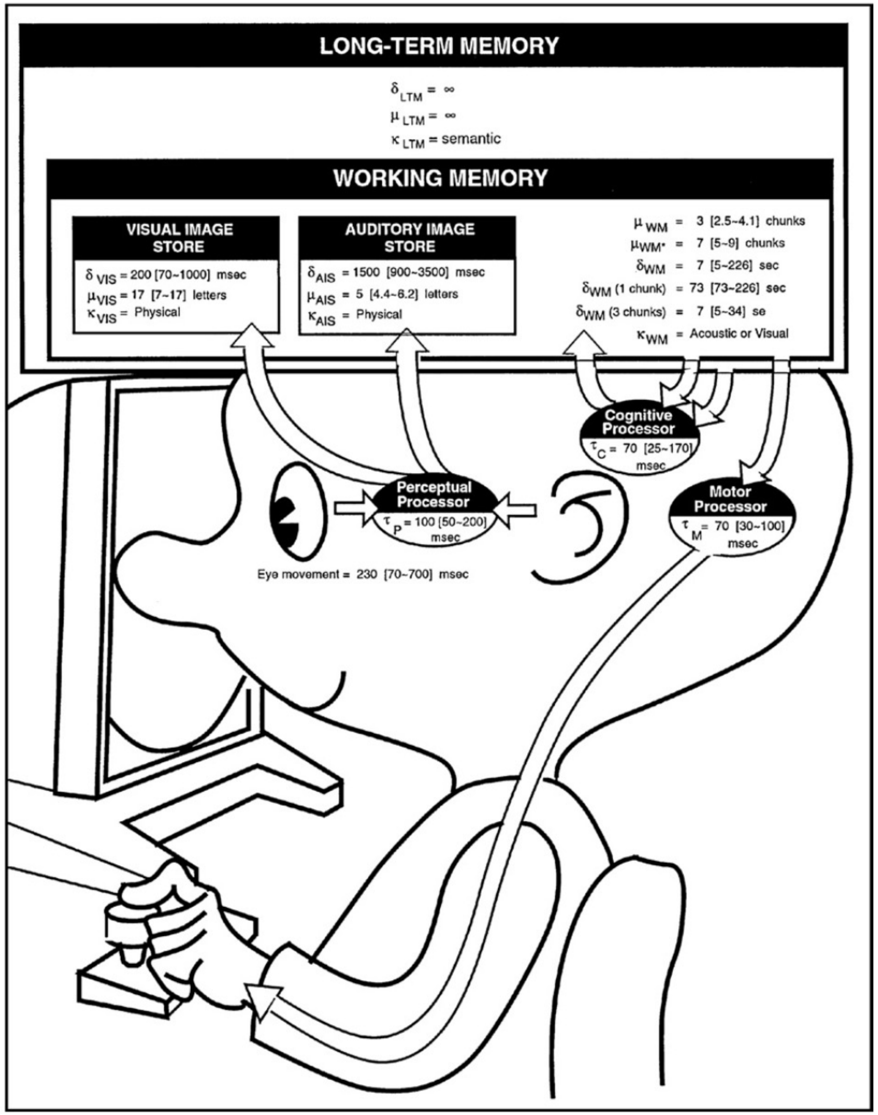


FIGURE 1.11

The model human processor (MHP) (Card et al., 1983, p. 26).

Human-Computer Interaction
Human Factors

Human Factors

Humans are more complex than machines.
Humans are also different!

HCI researchers must understand users' abilities and limitations.

2.1 Newell's time scales
2.2 Human Factors: sensor/responder vs display/control
2.3 Human sensors (vision, hearing, touch, smell, taste)
2.4 Responders (limbs, handedness, voice, eyes)
2.5 Perception
2.6 Language

Timing human action

Time is often a key variable in an HCI experiment

Scale (sec)	Time Units	System	World (theory)
10^7	Months		**SOCIAL BAND**
10^6	Weeks		
10^5	Days		
10^4	Hours	Task	**RATIONAL BAND**
10^3	10 min	Task	
10^2	Minutes	Task	
10^1	10 sec	Unit task	**COGNITIVE BAND**
10^0	1 sec	Operations	
10^{-1}	100 ms	Deliberate act	
10^{-2}	10 ms	Neural circuit	**BIOLOGICAL BAND**
10^{-3}	1 ms	Neuron	
10^{-4}	100 µs	Organelle	

FIGURE 2.1

Newell's time scale of human action.

(From Newell, 1990, p. 122)

workplace habits, groupware usage patterns, social networking, online dating, privacy, user styles and preferences, etc.

web navigation, user search strategies, collaborative computing, social navigation, and situated awareness

selection techniques, menu design, auditory feedback, text entry, gestural input, …

My research

Human models …

Useful for studying human brain, sensors, and responders

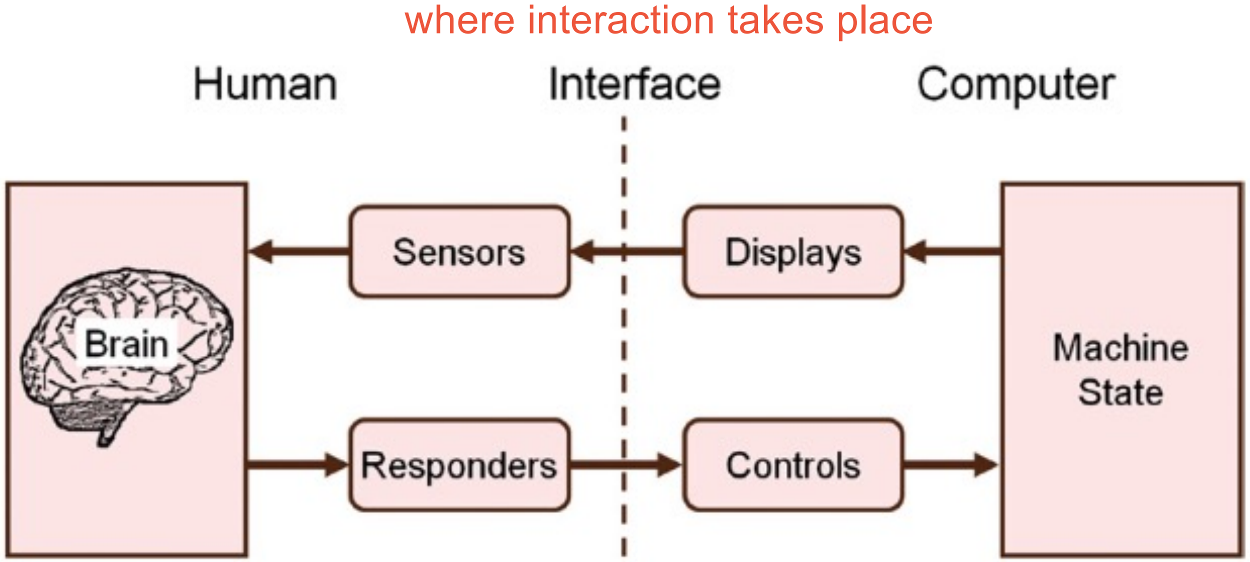


FIGURE 2.2

Human factors view of the human operator in a work environment.

(After Kantowitz and Sorkin, 1983, p. 4)

Human models …

Useful for studying human brain, sensors, and responders

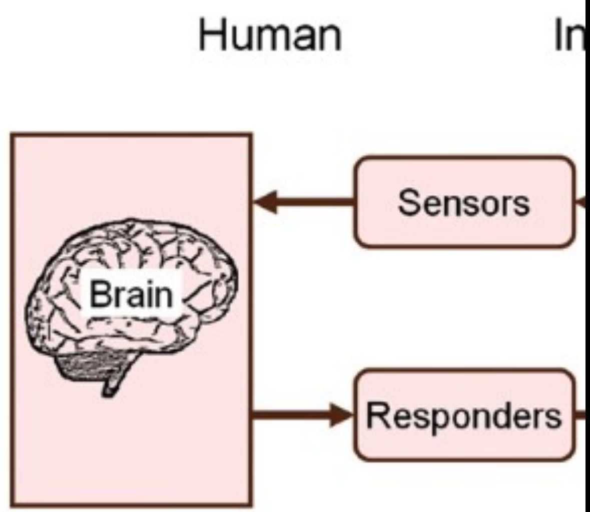


FIGURE 2.2

Human factors view of the human operator

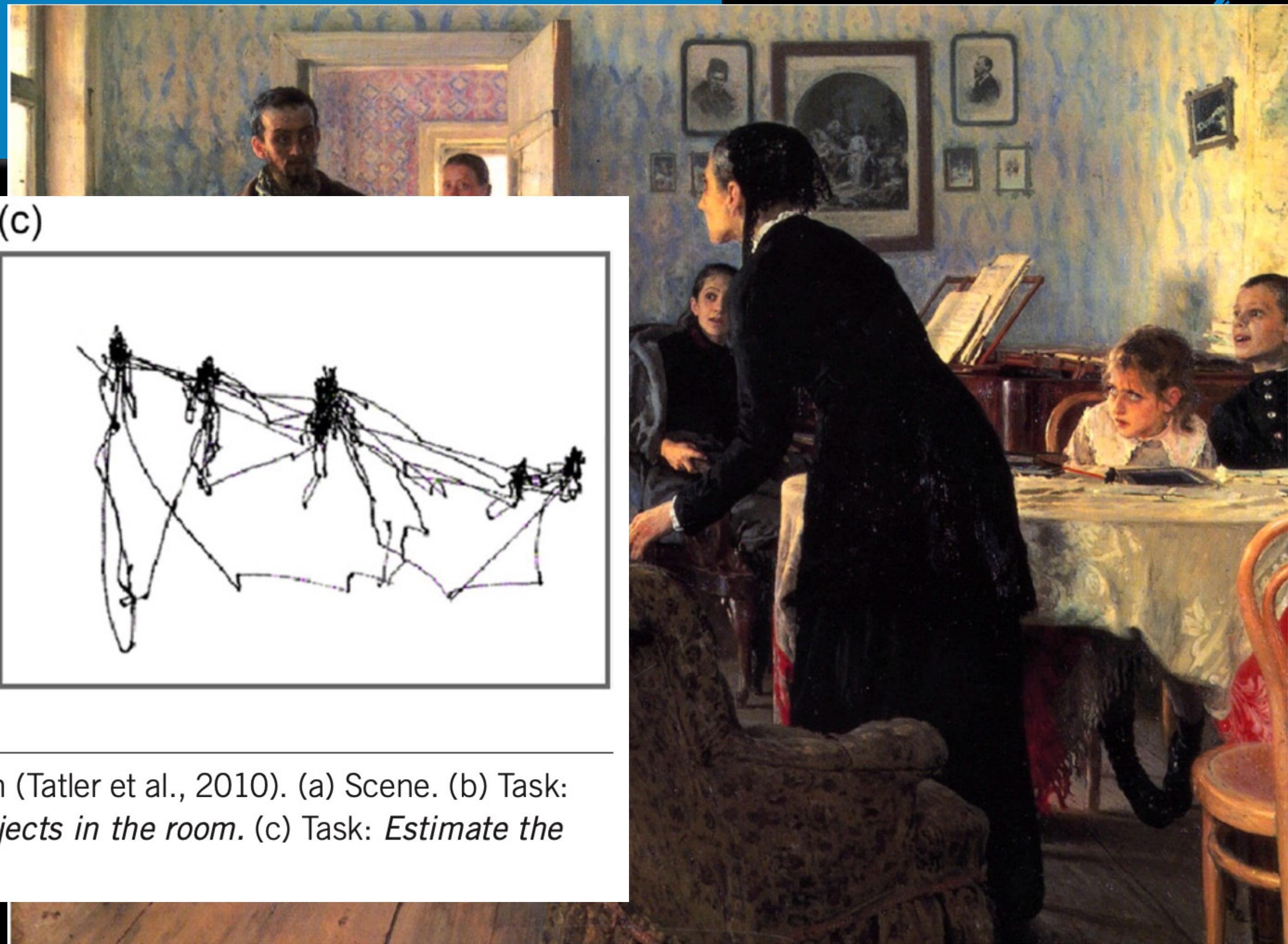
Include vision (sight), hearing (audition), touch (tactician), Smell (olfaction),Taste (gustation), other?

Include limbs (hands, fingers, feet, leg, head, etc.), voice, eyes (using eye tracker)

Sensors: vision / sight

- In many interaction scenarios the (most) important sense
- Often not passive: eyes move in fast series of fixations (200ms) and saccades (30-120ms)
- Eye tracking often used in HCI studies

Eye scan paths



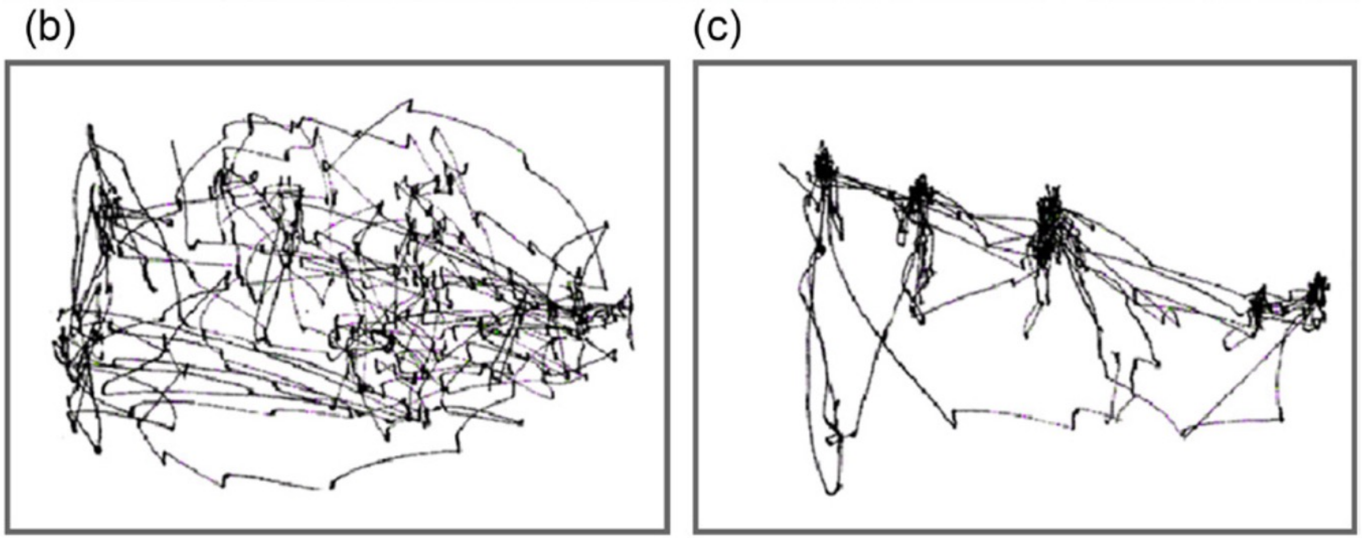


FIGURE 2.6

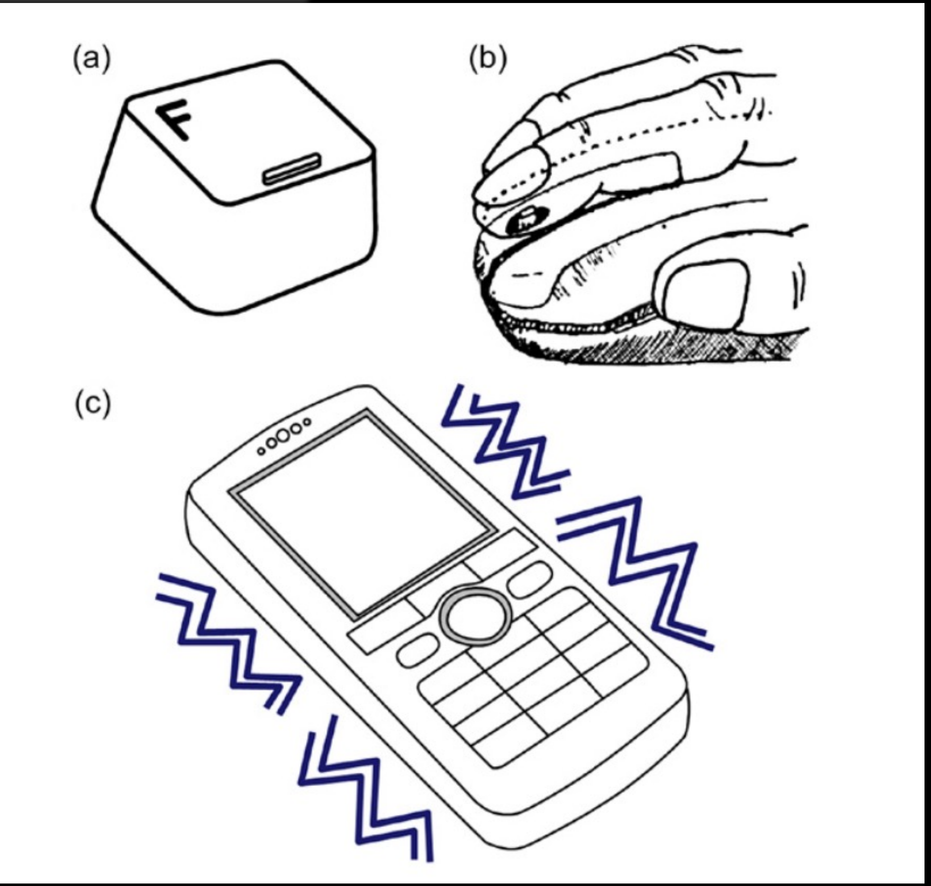
Yarbus' research on eye movements and vision (Tatler et al., 2010). (a) Scene. (b) Task: *Remember the position of the people and objects in the room.* (c) Task: *Estimate the ages of the people.*

Sensors: hearing/audition

- Increasingly important
- Devices we interact with are also used for listening to music
- Speech recognition & speech generation
- Accessibility when visual interfaces are not suited

Sensors: touch / tactition

- (passive) identifiers on keyboards (F/J)
- Edge, shape and resistance of keys, mouse buttons, joysticks
- (active) vibration phones, game controllers



Responders

- Limbs, mainly hands
  - Design w.r.t. left/right handedness
- Voice
- Eyes

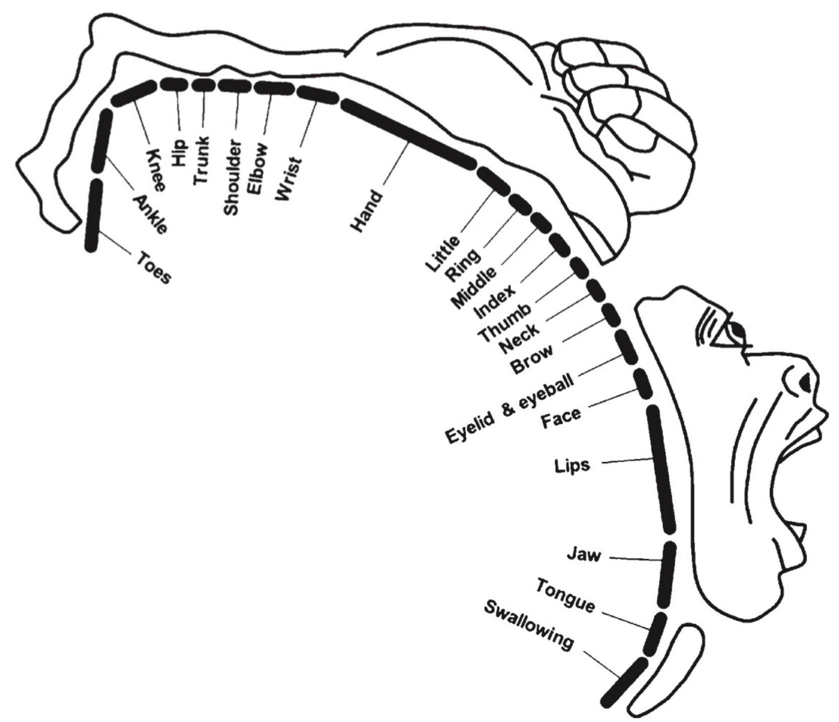


FIGURE 2.9

Motor homunculus showing human responders and the corresponding cortical area.

(Adapted from Penfield and Rasmussen, 1990)

Eye typing:
eye as sensor **and** responder

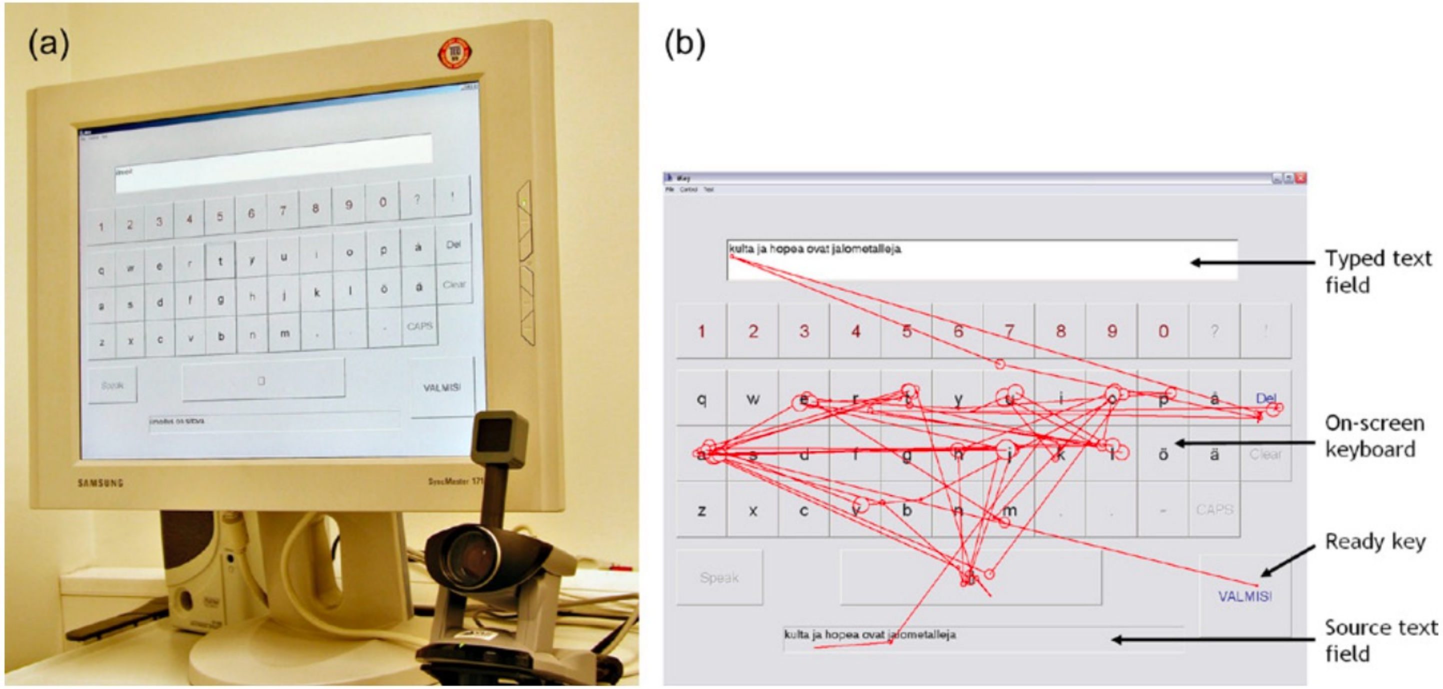


FIGURE 2.13

Eye typing: (a) Apparatus. (b) Example sequence of fixations and saccades (Majaranta et al., 2006).

Brain

- Perception
  - Associations & meaning
  - Just noticeable difference (JND)
  - Ambiguity/illusions
- Cognition
  - Plays often key role in task of HCI experiment
  - Thinking takes time too!
- Memory
  - Long/short term
  - The Magic Number Seven, Plus or Minus Two

- Control
  - Brain-computer interfaces

Language

- Spoken vs written
- Language differences
- Redundancy & entropy

(a) My smmr hols wr CWOT. B4, we used 2go2 NY 2C my bro, his GF & thr 3 :- kids FTF. ILNY, it's a gr8 plc.

(b) My summer holidays were a complete waste of time. Before, we used to go to New York to see my brother, his girlfriend and their three screaming kids face to face. I love New York. It's a great place.

FIGURE 2.21

Shortening English: (a) SMS shorthand. (b) Standard English.

"In HCI, our interest in language is primarily in systems of writing and in the technology that enables communication in a written form."

Human performance

Humans use their sensors, brain, and responders to do things. When the three elements work together to achieve a goal, human performance arises.

- Reaction time
- Human Error
- Attention
- Practice & skill